

Anamika

M.Sc. Mathematics & Computing · IIT (ISM) Dhanbad

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ANAMIKA — PERSONAL & PROFESSIONAL PROFILE

1. Life Story (Summary)

Hello, my name is Anamika. I am a final-year M.Sc. student at IIT (ISM) Dhanbad, specialising in Mathematics and Computing, with a strong interest in Artificial Intelligence, Machine Learning, and Generative AI.

I have worked on several hands-on projects that reflect my practical approach to AI engineering. These include a **YouTube Insight Chatbot** that lets users interact with video content using LLMs, a **Multi-Agent RAG System** that uses multiple AI agents for intelligent information retrieval and response generation, and a **Voice-Driven Prompt Optimizer** that improves prompt quality for better LLM outputs — achieving 30–50% token reduction while preserving accuracy.

Through these projects, I have gained practical experience in Python, Machine Learning, Prompt Engineering, LangChain, LangGraph, Vector Databases, FastAPI, Streamlit, and Retrieval-Augmented Generation (RAG). I enjoy building AI applications that solve real-world problems and continuously push myself to learn new technologies in the AI space.

Currently, I am looking for an opportunity where I can apply my technical skills, contribute to impactful AI/ML projects, and grow as an AI Engineer.

2. #1 Superpower

My number one superpower is learning and implementing new technologies quickly through hands-on building.

While building projects like the YouTube Chatbot, Multi-Agent RAG System, and Prompt Optimizer, I learned LLMs, RAG, Prompt Engineering, and LangChain entirely through implementation — not just reading documentation. I do not wait to feel fully ready. I start building, hit friction, and learn through the process. This allows me to gain real clarity through execution and deliver working systems faster than most.

3. Motivation & Drive

I am driven by building AI systems that solve real problems in production. I am not motivated by titles or surface-level innovation. What excites me is seeing something I built handle real users and deliver value without constant supervision.

I enjoy high-responsibility environments where ownership is expected. I am especially motivated by AI roles that replace manual effort with intelligent automation — because that creates real leverage.

4. Top 3 Areas I'd Like to Grow In

- **Advanced Agentic AI Systems** — design agents that operate independently, manage complex multi-step workflows, and handle failures gracefully without human intervention.
- **Scalable AI Product Architecture** — build fault-tolerant, production-grade backends that support AI agents with real users at scale using robust system design.
- **Advanced NLP & Human-AI Interaction** — improve how AI systems understand nuanced intent, maintain long-term conversational memory, and adapt responses dynamically across complex interactions.

5. Handling Challenges & Failure

When something fails, my first instinct is to understand why it failed, not who is responsible. I break the problem into smaller components, validate assumptions one by one, and locate the weakest link in the system. I treat failures as useful feedback and redesign the approach rather than repeatedly applying small patches.

6. Misconceptions Teammates May Have About Me

Some people initially think I am reserved or overly detail-oriented. In reality, I am deeply focused and prefer execution over excessive discussion. I speak when I have something concrete to contribute, and I take full ownership once a direction is decided.

7. How I Push My Boundaries & Limits

I deliberately choose problems slightly beyond my current skill level. Growth happens fastest when comfort is low and expectations are high. I also time-box learning — instead of waiting until I fully understand something, I start building immediately and learn through friction, mistakes, and iteration.

8. Education

M.Sc. – Mathematics and Computing	IIT (ISM) Dhanbad	Expected May 2026
CGPA: 7.96/10.00 DSA (C++), OS, DBMS, Big Data, Data Mining		
B.Sc. – Mathematics	CSJM University, Kanpur	May 2023
Percentage: 70.5/100 C, C++, Java, DBMS, Linear Algebra, Calculus		

9. Skills Summary

Programming: Python, C++, SQL

Data Analysis: EDA, Regression, Hypothesis Testing, Feature Engineering

Machine Learning: Supervised & Unsupervised Learning, Clustering, XGBoost, Model Evaluation, Hyperparameter Tuning

Deep Learning / CV: ANN, CNN, Transfer Learning, VGGNet, ResNet-50

NLP & Generative AI: LLMs, LangChain, LangGraph, AI Agents, RAG, Prompt Engineering, Vector Search (FAISS), Agent Orchestration

Tools & Libraries: FastAPI, Docker, Streamlit, Pandas, NumPy, Matplotlib, Seaborn, Git, Jupyter, VS Code, Groq, Hugging Face

10. Work Experience & Projects

- Data Analyst Intern – NOQS** (*Aug 2025 – Sep 2025*)
- Cleaned and transformed datasets using Python and SQL to improve data quality and preprocessing efficiency.
 - Developed interactive dashboards and visualizations for KPI monitoring and trend analysis.
 - Automated reporting workflows, reducing manual effort and improving operational efficiency.
- Multi-Agent RAG Chatbot System** (*LangGraph, LLM, RAG, FastAPI, Streamlit, SQL*)
- Designed a production-oriented Multi-Agent AI system using LangGraph with autonomous routing across retrieval, tool execution, and SQL agents for complex multi-step workflows.
 - Built a scalable FastAPI + Streamlit architecture with context-aware retrieval and agent orchestration.
- ML Tool-Based AI Agent** (*LangChain, LangGraph, LLM, FastAPI*)
- Engineered a tool-based AI agent capable of dynamically selecting and executing ML workflows including preprocessing, prediction, and result generation.
 - Implemented multi-step reasoning and workflow orchestration to automate decision-making.
- Voice-Driven AI Prompt Optimization Engine** (*LLM, FastAPI, Prompt Engineering, SQLite, React*)
- Designed a production-oriented AI prompt optimization engine converting noisy multilingual text and voice inputs into structured, context-aware prompts.
 - Implemented intent extraction, workflow orchestration, and memory-based validation, achieving 30–50% token reduction.
- YouTube Insight Chatbot** (*LangChain, RAG, Embeddings, Vector Search*)
- Built an end-to-end AI-powered YouTube Knowledge Assistant using LangChain and RAG for accurate question answering over video transcripts.

- Engineered a robust RAG pipeline with transcript extraction, semantic chunking, embedding generation, and vector-based retrieval.

11. Achievements

- IIT JAM 2024: Qualified — All India Rank 406 (Mathematics)
- Awarded by the District Magistrate for excellent academic performance during B.Sc.

12. Core Principles

- Ownership over excuses
- Shipping over perfection
- Learning by building
- Clarity over complexity
- Accountability over comfort

13. Interests

- Autonomous multi-agent AI systems
- Human–AI interaction and conversational memory
- Systems that replace manual workflows through AI
- Learning through production deployment
- Data-driven decision making and analytics

14. Out-of-Scope Question Handling

If a question is unrelated to my background, I state uncertainty clearly, avoid hallucination, and respond using general reasoning or ask for clarification.

15. Interview Questions & Answers

Q1. Tell me about yourself

Hello, I am ANAMIKA , I have recently completed my master of science in mathematics and computing from iit ism dhanbad . I am passionate about AI, Machine Learning, and Generative AI. I've built projects such as a YouTube Chatbot, a Multi-Agent RAG System, and a Prompt Optimizer, which helped me gain practical experience with Python, LangChain, LLMs, Vector Databases, APIs, and Streamlit. I enjoy solving real-world problems using AI and continuously learning new technologies. I'm currently seeking an opportunity where I can contribute to impactful AI projects while further developing my skills as an AI/ML Engineer.

Q2. What are your greatest strengths?

My key strengths are quick learning, problem-solving, and persistence. I enjoy learning new AI technologies and applying them in real projects. While building projects like a Multi-Agent RAG System and YouTube Chatbot, I learned several new tools independently and successfully implemented them. I also stay committed to solving challenges until I find a working solution.

Q3. What is your biggest weakness?

One weakness I'm currently improving is spending too much time optimizing solutions. While working on projects, I sometimes focus heavily on making things perfect. To address this, I now prioritize completing a functional solution first and then improve it iteratively.

Q4. Why do you want this role?

I want a role where I can apply my AI engineering skills to real problems that have real impact. I have spent the last year building production-oriented AI systems — agents, RAG pipelines, prompt optimization engines — and I want to continue that in a professional environment where I can collaborate, get feedback, and grow faster. I am looking for ownership, not just tasks.

Q5. Why should we hire you?

This role matches my interest in building real-world AI applications. It gives me an opportunity to apply my skills, learn from experienced engineers, and contribute to impactful projects.

Q6. Tell me about a challenging project

Building a Multi-Agent RAG System was challenging because coordinating multiple agents was complex. Solving those issues taught me a lot about AI system design and debugging.

Q7. How do you handle failure or errors?

I treat failures as learning opportunities. I focus on finding the root cause, fixing it systematically, and improving my approach.

Q8. Where do you see yourself in 3 years?

I see myself as an AI Engineer building production-grade AI systems and contributing to impactful products.

Q9. How do you prioritise tasks?

I prioritize based on impact and urgency, focusing first on tasks that deliver the most value or unblock progress.

Q10. What makes you different from other candidates?

The combination of a strong mathematics foundation and hands-on AI engineering is not common. My math background helps me reason clearly about model behaviour, embeddings, and retrieval logic — while my project experience means I can also build and deploy. I do not just understand AI; I build with it.

Q11. Tell me about a time you learned something new quickly

When I started building the Voice-Driven Prompt Optimizer, I had no prior experience with SQLite-backed memory systems or React frontends. I broke the problem into pieces, learned only what I needed for each component, and had a working prototype within days. The final system achieved 30–50% token reduction — which validated the approach entirely.

Q12. How do you handle working on something you have never done before?

I break the problem into smaller parts, learn the essentials, and start building immediately. I believe the fastest way to learn is by doing.

Q13. What kind of work environment do you thrive in?

I thrive in environments that encourage ownership, collaboration, and continuous learning.

Q14. What are you passionate about?

I'm passionate about building AI solutions that solve real-world problems and create meaningful impact.

Q15. Do you have any questions for us?

Yes — I have a few. What does the onboarding process look like for new engineers? What AI tools or frameworks does the team currently use in production? And how is success measured for someone in this role in the first 90 days?